Torque-Ripple Reduction in PM Synchronous Motor Drives Using Repetitive Current Control

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***Abstract*—The paper deals with the torque-ripple reduction in a permanent magnet synchronous motor (PMSM) drive with distorted back electromotive force. A smooth torque is obtained by tracking a modified current reference which is periodic over one-sixth of the electrical time period in the synchronous reference frame. An accurate tracking involves, however, very high current loop bandwidth, which is usually not achievable with conventional linear controllers. In order to improve current tracking in the presence of periodic reference signals and disturbances, the paper proposes the application of repetitive techniques to the current control in a field-oriented PMSM drive, where the q-axis current reference has been modified to achieve constant torque. The paper investigates the advantages and pitfalls of the method, through a mathematical analysis and an experimental validation obtained on a laboratory prototype. Particular emphasis is placed on the adjustments that have been specifically studied to enhance the overall system performance.**

***Index Terms*—Permanent magnet synchronous motor (PMSM) drives, repetitive control, torque-ripple.**

I. Introduction

PERMANENT magnet synchronous motor (PMSM) drives are rapidly gaining market shares over the traditional competitors. The rapid growth, as usual, deals with cost, efficiency, and reliability. PMSM drives also feature one of the highest torque-to-losses ratio, for the sake of energy saving and environmental care. PMSM drives find their natural application in fast dynamic positioning systems and machine–tool spindle drives. To obtain a constant torque, under the hypothesis of pure sinusoidal back electromagnetic force (EMF), the conventional field oriented control (FOC) imposes constant current references in the synchronous reference frame. Nevertheless, depending on the magnet shapes and the windings manufacture, the back-EMF has in practice very different shapes, which range from almost sinusoidal to trapezoidal ones. The result is twofold: a torque-ripple generation if a constant reference is maintained, and the requirement of more sophisticated current control loops if the current reference is properly modified [1], [2]. The cost of fixed-point digital signal processors (DSPs) is decreasing to unseen levels, so that the implementation of control strategies different from classical proportional–integral (PI) regulators can be profitably undertaken, wherever PI controllers exhibit bandwidth limits.

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Active compensation of torque harmonics implies high current control bandwidth. A possible solution is represented by deadbeat current control [1], merged with predictive algorithms to reduce system delays, as far as possible. The torque-ripple is reduced to a few percent, at the cost of increased control complexity due to feedforward compensators of the flux-linkage harmonics that have to be tuned for the specific motor. Other fast current control techniques can be used for torque-ripple reduction, such as the hysteresis current control, and some examples are reported in [3] and [4].

Instead of using a high current control bandwidth, which usually requires a tradeoff among performance, stability and complexity, and taking into account that the modified current references for torque-ripple reduction are inherently periodic in the synchronous reference frame, this paper proposes the use of repetitive control techniques, which have been shown to be an effective method for tracking period references and compensating periodic disturbances [5]–[10].

The rather specialized subject of repetitive control was first investigated around the beginning of the 80's. Perhaps boosted by the use of computers in control applications, the ability to store a whole period of the reference or estimated disturbance signal made possible the practical application of these techniques. A typical example of repetitive control application is the rejection of disturbances acting on the track-following servo system of optical disk drives [11], [12]. The control loop inherently contains significant periodic components, linked to deficiencies in track geometry and eccentric rotation of the disk, which cause periodic tracking errors. Such disturbances have been effectively rejected by employing a repetitive controller. Other significant applications have been reported for pulsewidth modulated (PWM) inverters used for uninterruptible power supplies (UPS), active filters and high-quality rectifiers, where the repetitive control has been used to minimize the periodic distortions resulting form nonlinear cyclic loads [13]–[17].

In this paper, the repetitive control technique is applied to the torque-ripple reduction in high-performance PMSM drives, where the q-axis current reference has been modified so as to impose a constant torque in spite of back-EMF distortion. More specifically, the repetitive controller is merged with a conventional PI controller, where the PI control dominates during transients and large-signal dynamics, while the repetitive control ensures the compensation of the remaining errors so as to achieve a near perfect tracking of a periodic current reference signal. The target is a smooth torque production, devoid of harmonics related to nonsinusoidal back-EMF. The paper is organizes as follows. Section II reports the motor model and the mathematical details of current reference generation. Section III contains the basics of repetitive control, together with the design criteria and the stability analysis of the proposed control scheme, while Section IV shows the hardware description and the experimental results. Section V reports some further design issues, mainly focused on the resolution of the speed measurement and the sampling time adaptation required to maintain the performance of the repetitive control under varying motor speed.

II. Current Reference Generation

The PMSM model is written in a synchronous (d, q) reference frame, fixed to the rotor flux linkage position ϑ_{me}, as reported hereafter

$$\begin{aligned} u_d &= R_s i_d + L_s \frac{di_d}{dt} - \omega_{\mathrm{me}} L_s i_q + e_d \\ u_q &= R_s i_q + L_s \frac{di_q}{dt} + \omega_{\mathrm{me}} L_s i_d + e_q \end{aligned} \tag{1}$$

where ω_{me} is the electrical speed, related to the mechanical one by $\omega_{\mathrm{me}} = p\omega_m$, being p the number of motor poles pairs. The phase back-EMFs e_a, e_b, e_c are measured offline, and transposed into the synchronous frame quantities $e_d, e_q, (e_0)$ by the Park transformation matrix. Fig. 1 reports the behavior of the phase a back-EMF, relative to a unitary electrical speed, for the PMSMs used in the experiments, whose data are reported in Table I. Possible parameter variations of the back-EMF due to motor temperature variations usually do not give significant contribution and, if needed, can be compensated by a measure of the rotor temperature or, equivalently, its estimation from an accurate thermal network.

Fig. 2 reports the back-EMFs e_d and e_q in the d–q synchronous reference frame for the same motor. While the fundamental components of e_a, e_b, e_c are mapped to a constant e_q voltage, the n-order harmonics $e_{a,n}(\vartheta_{\mathrm{me}}) = -E_n \sin(n\vartheta_{\mathrm{me}} + \varphi_n), e_{b,n}, e_{c,n}$ are mapped by the *abc/dq* Park transformation to $(n-1)$-order and $(n+1)$-order harmonics, according to the following expression:

$$\begin{aligned} e_{q,n}(\vartheta_{\mathrm{me}}) = \frac{1}{3} E_n \Bigg[& \cos((n-1)\vartheta_{\mathrm{me}} + \varphi_n) \\ & - \cos((n+1)\vartheta_{\mathrm{me}} + \varphi_n) \\ & + \cos\left((n-1)\left(\vartheta_{\mathrm{me}} - \frac{2\pi}{3}\right) + \varphi_n\right) \\ & - \cos\left((n+1)\left(\vartheta_{\mathrm{me}} - \frac{2\pi}{3}\right) + \varphi_n\right) \\ & + \cos\left((n-1)\left(\vartheta_{\mathrm{me}} - \frac{4\pi}{3}\right) + \varphi_n\right) \\ & - \cos\left((n+1)\left(\vartheta_{\mathrm{me}} - \frac{4\pi}{3}\right) + \varphi_n\right)\Bigg] \end{aligned} \tag{2}$$

where the electrical position ϑ_{me} is the time integral of the electrical speed and E_n and φ_n are the amplitude and the initial phase of the nth order harmonic, respectively.

Since e_a is almost a pure odd function, the first relevant harmonic is the fifth-order one, which produces a sixth-order dominant harmonic in the d–q reference frame, as shown in Fig. 2.

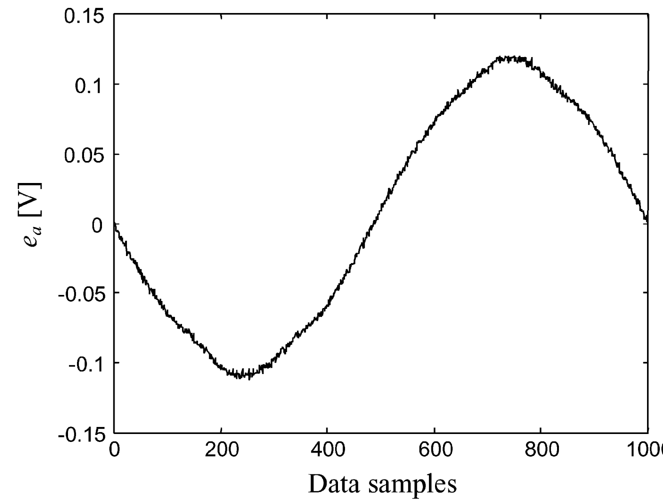

Fig. 1. Phase a back-EMF at $\omega_{\mathrm{me}} = 1$ rad.el/s.

TABLE I
NAMEPLATE DATA AND PARAMETERS OF THE PMSM

Description	Value	Symbol
Nominal torque	2.5 Nm	T_n
Nominal speed	3000 rpm	Ω_n
Phase voltage	196 V	U_n
Rated Current	4.4 A	I_n
PM Flux Linkage	0.1275 Vs/rad	Λ_{mg}
Phase Resistance	1.29 Ω	R_s
Phase Inductance	7 mH	L_s
Pole Pairs	3	p

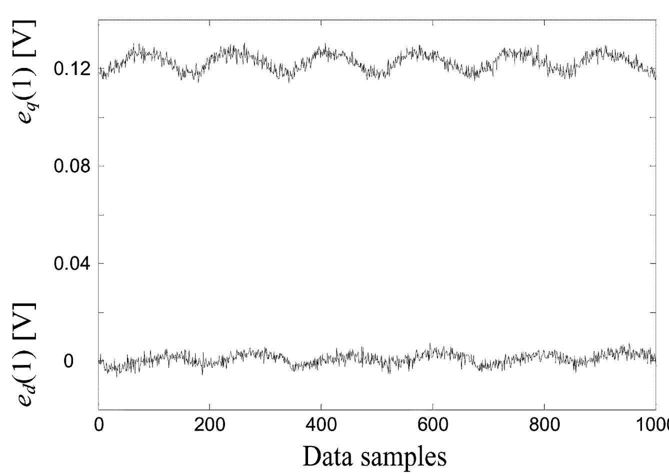

Fig. 2. Back-EMF at $\omega_{\mathrm{me}} = 1$ rad.el/s in the d–q reference frame.

By equating the mechanical and the net electrical power, the following expression is obtained:

$$\frac{3}{2}(e_d i_d + e_q i_q + e_o i_o) = \omega_m \tau \tag{3}$$

where i_d, i_q, i_o represent the phase currents transformed in the synchronous d, q reference frame.

Since the motor is without neutral connection, the homopolar current i_o is null. In principle, nonsinusoidal back-EMFs lead to $e_d, e_q \neq 0$. In practice, as shown in Fig. 2, e_d is quite small and, without significant loss of efficiency in the torque production, it is possible to set $i_d^* = 0$. Thus, a simple and straightforward derivation of the q-axis current component i_q^* from the torque reference is obtained

$$i_q^* = \frac{2}{3}\tau^* \frac{\omega_{\rm me}}{p}\frac{1}{e_q} = \frac{2}{3}\tau^* \frac{\omega_{\rm me}}{p}\frac{1}{e_{q(1)}\omega_{\rm me}} = \frac{2}{3p}\frac{\tau^*}{e_{q(1)}} \tag{4}$$

where $e_{q(1)}$ is the q-axis component of the transformed back-EMF vector, referred to $\omega_{\rm me} = 1$ rad·el/s. In order to meet a constant torque request, and due to the tight link to the back-EMF e_q, the q-axis current reference i_q^* contains an alternating component, which has the fundamental frequency at six times the electrical frequency. It is also worthwhile to eliminate the high-frequency components, which come from the sampled back-EMF waveforms, and are definitely of no use. The current reference can thus be obtained as the sum of a dc term $(i_{q\rm dc}^*)$ and an alternating one $(i_{q\rm ac}^*)$, i.e.,

$$i_q^* = i_{q\rm dc}^* + i_{q\rm ac}^* = \left(K_{\rm tau} + i_{q(1)\rm ac}\right)\,\tau^* \tag{5}$$

where $i_{q(1)\rm ac}$ is the alternating reference relative to a constant unitary torque. Fig. 3 shows the q-axis current reference (i_q^*) for the motor used during tests. The high-order harmonics (greater than twelve) have been removed from the reference calculated by (4).

The complete schematic of the current reference generator is reported in Fig. 4, [2]. The torque reference τ^* is supplied by the speed regulator PI_ω, which processes the speed error (ω_m^*-ω_m). The dc-component $i_{q\rm dc}^*$ is directly obtained by multiplying τ^* by

$$K_{\rm tau} = \frac{1}{2\pi}\int_0^{2\pi} i_q^*(\vartheta_{\rm me})d\vartheta_{\rm me}\Bigg|_{\tau*=1Nm} \tag{6}$$

where i_q^* is evaluated for $\tau^* = 1$ (Fig. 3), while $i_{q\rm ac}*$ is the product of $\tau*$ by the normalized alternate reference (Fig. 3), which is stored in a look-up table (LUT). The actual value is retrieved from the DSP memory, by using the measured electrical position $\vartheta_{\rm me}$ as a pointer.

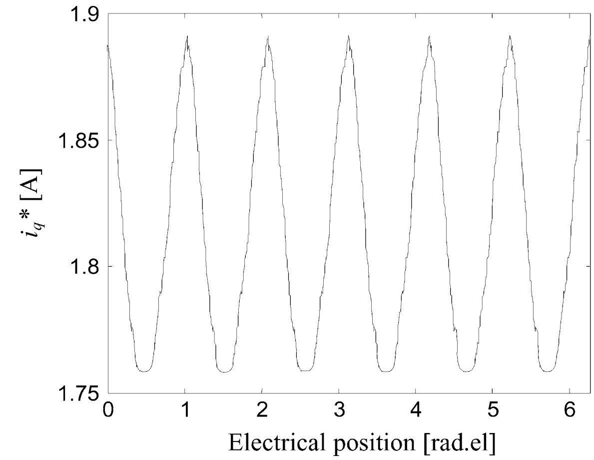

Fig. 3. q-axis current reference at $\tau^* = 1$ Nm.

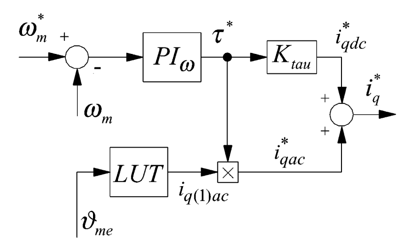

Fig. 4. Current reference generator.

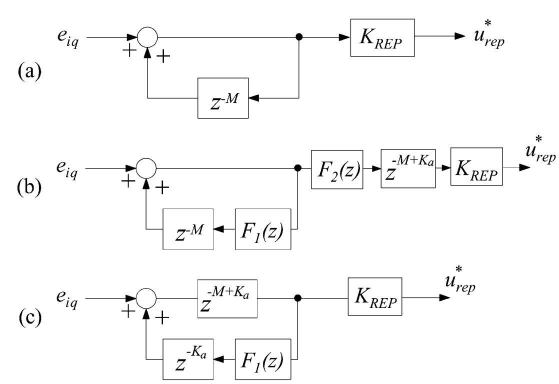

Fig. 5. (a) General implementation of the repetitive-based control, (b) provisions for improved stability margin, and (c) the adopted solution.

III. Proposed Repetitive Current Control

A. Basics of Repetitive Control

The basic structure of the proposed repetitive controller is shown in Fig. 5. The concept of repetitive control theory originates from the internal model principle [3], so that the controlled output tracks a set of reference inputs without steady-state errors if the model that generates these references is included in the stable closed-loop system.

For example, if the system is required to have a zero steady-state error to sinusoidal input, then the model of the sinusoidal function (i.e., $\omega_{\rm me}^2/(s^2+\omega_{\rm me}^2)$, being $\omega_{\rm me}$ the corresponding angular frequency) should be included in the loop gain. In order to implement a repetitive control system, a periodical reference must be generated. Its digital implementation includes simply a delay line and positive feedback [14], as shown in Fig. 5(a).

In practice, however, this scheme usually leads to instability since it amplifies many high-order harmonics (theoretically up to the Nyquist frequency), while the system to be controlled has a limited bandwidth so that the elimination of all harmonics is unfeasible. Indeed, the provisions needed to take into account the practical constraints of the process to be controlled become one of the most important factors in the design of repetitive controllers. For example, in order to guarantee system stability, some filters need to be introduced in the scheme of Fig. 5(a), either in the feedback path $(F_1(z))$ or in cascade to the repetitive control $(F_2(z))$ or using both $F_1(z)$ and $F_2(z)$, as shown in Fig. 5(b). Moreover, stability of the repetitive controller can be improved by adding a delay of M–K_a samples at the output of the regulator, which is equivalent to a leading time of K_a samples for the periodic frequencies [16]. The leading action does not change the gain at the specified frequencies but enforces the system phase margin. The solution adopted in this paper is shown in Fig. 5(c), which is theoretically equivalent to the scheme of Fig. 5(b) with $F_2(z) = 1$.

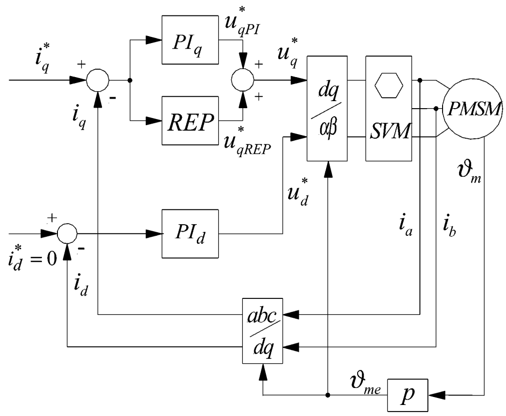

Fig. 6. Proposed drive for the speed control of a PMSM.

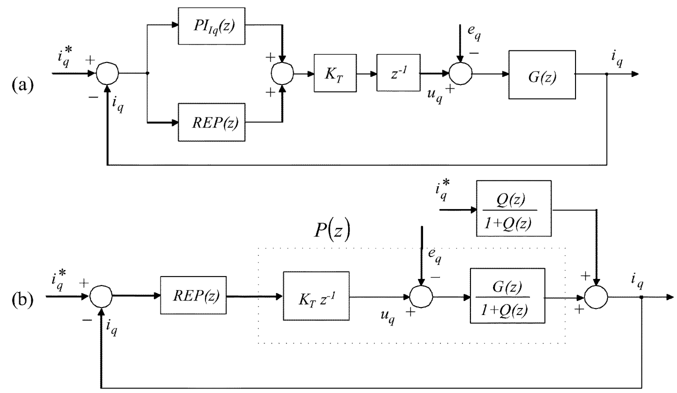

Fig. 7. (a) Block diagram of the q-axis current control under the assumption of ideal decoupling and (b) equivalent representation.

B. Proposed Discrete-Time Control System

Fig. 6 reports the block diagram of the proposed control structure. A standard PI regulator and the proposed repetitive (REP) controller are used in parallel, and the sum of their output form the actual q-axis voltage reference $u_q^*(u_q^* = u_{q\mathrm{PI}}^* + u_{q\mathrm{REP}}^*)$. Due to the reduced influence of i_d current in the torque production, a PI controller alone is maintained for the d-axis.

At steady-state, the tracking error is zero, the output of the q-axis PI control $(u_{q\mathrm{PI}}^*)$ is constant and the output of the repetitive control $(u_{q\mathrm{REP}}^*)$ is a periodic signal. Several experiments have confirmed that this topology is particularly suited for an effective digital implementation, as discussed hereafter. While the PI action gives the most relevant contribution in the generation of the voltage reference (u_q*), the repetitive controller is only aimed to the reduction of the high-frequency periodic current errors, which the PI control is not able to compensate due to its limited bandwidth. The contribution of the repetitive control, while being essential for the torque-ripple compensation, remains anyway within a few percent of the total voltage reference. Moreover, during transient conditions, the output of the repetitive control does not change significantly, due to the absence of periodic excitation. Thus, the antiwind-up algorithm for the repetitive part of the controller was simply implemented using a saturation of the output block $z^{-(\mathrm{M-Ka})}$ of Fig. 5(c); we found this provision sufficient in order to avoid undesired transients after inverter saturations. It has been experienced that this distribution of tasks between the PI and the repetitive controllers yields better responses to large signal transients, with respect to the repetitive regulator alone. Further improvements on the stability and the transient response have been obtained by adopting the structure reported in Fig. 5(c), with a transfer function given by

$$\mathrm{REP}(z) = K_{\mathrm{REP}} \frac{z^{-M+K_a}}{1 - z^{-M} F_1(z)} \tag{7}$$

where K_{REP} is the controller gain and M is the ratio between the period $T_{\mathrm{rep}} = 2\pi/\omega_{\mathrm{rep}}$ of the alternating component of the q-axis current reference (Fig. 3) and the sampling period T_c. Since M must be integer, the sampling time must be an exact sub-multiple of period T_{rep}. The following relation links the mechanical speed ω_m, T_c, and M

$$\omega_m = \frac{\omega_{\mathrm{rep}}}{hp} = \frac{2\pi}{hpT_{\mathrm{rep}}} = \frac{2\pi}{hpMT_c} \tag{8}$$

where h is the order of the dominant harmonic component of the current reference with respect to the electrical position ω_{me} and, thus, $h = 6$ in the proposed application. $F_1(z)$ includes a low-pass moving average filter with three taps and a high-pass filter. The former improves the stability margin, by reducing the repetitive control gain at high frequency, while presenting a constant delay of T_c, which can be easily compensated by setting $M-1$ instead of M in the denominator of (7). The high-pass filter limits the dc gain of the repetitive control which otherwise would be infinite. Since both the PI and the repetitive control without the high-pass filter behave as integrators at low frequency, this provision avoids the fact that the dc component of both regulators evolves in a uncontroller manner.

C. Design Criteria and Stability Analysis

In order to understand stability limits of the proposed solution, let us focus on the q-axis current control. Under the assumption of ideal decoupling between the d and q axis, which is obtained by adding to the output of the q-axis regulator a term equal to $\omega_{\mathrm{me}} L_s i_d$, the equivalent block diagram shown in Fig. 7(a) can be easily derived, where $G(z)$ is the transfer function which accounts for the R_s–L_s admittance, K_T is the static gain representing the space vector modulation (SVM), and the unit delay (z^{-1}) includes the DSP computational delay.

To highlight the properties of the repetitive control, the block diagram of Fig. 7(a) can be rearranged as reported in Fig. 7(b), where $Q(z)$ is the loop gain of the current control without the repetitive block, i.e.,

$$Q(z) = K_T z^{-1}\, \mathrm{PI}_q(z) G(z). \tag{9}$$

From Fig. 5(c), Fig. 7, and the proposed repetitive control (7), the transfer function between the error on the q-axis current e_{iq} and the q-axis current reference i_q* is given by

$$\frac{e_{iq}(z)}{i_q^*(z)} = \frac{1}{1+Q(z)} \frac{z^M - F_1(z)}{z^M - \left[\underbrace{F_1(z) - K_{\mathrm{REP}} z^{K_a} P(z)}_{H(z)}\right]} \tag{10}$$

where $P(z)$ is the transfer function that follows the REP control block in Fig. 7(b), i.e., $P(z) = K_T z^{-1} G(z)/(1+Q(z))$. Since

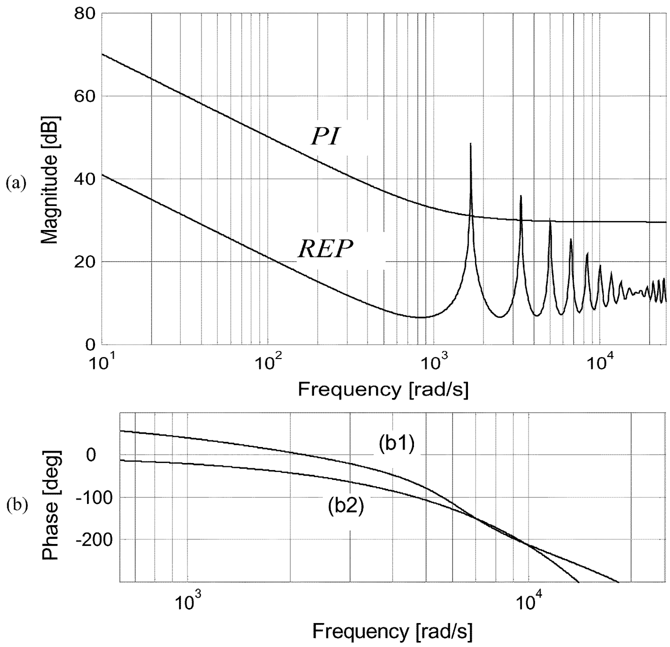

Fig. 8. (a) Bode diagram of $\mathrm{PI}(\exp(\mathrm{j}\omega T_c)$ and $\mathrm{REP}(\exp(\mathrm{j}\omega T_c))$; (b1) phase of $P(\exp(\mathrm{j}\omega T_c))$ and (b2) phase of $\exp(-\mathrm{j}3\omega T_c)$.

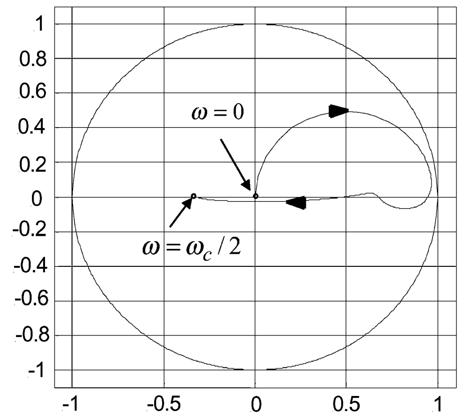

Fig. 9. Real (x-axis) and imaginary (y-axis) parts of $H(e^{\mathrm{j}\omega T_c})$.

the roots of the denominator in (10) are the closed-loop poles of the controlled system, stability analysis can be performed looking at the magnitude of the roots of the denominator of (10). If all the roots have a magnitude less than one, then all poles lie within the unity circle and the system is stable. The poles of function $1/(1+Q(z))$ coincide with those of the PI current control alone. By assuming that they yield a stable control system, the overall stability analysis can be performed looking at the remaining part of the denominator of (10). A sufficient condition for the system stability is [16]

$$|H(z)|_{z=e^{j\omega T_c}} < 1 \tag{11}$$

where ω increases from zero up to the Nyquist angular frequency.

1) Design of Parameter K_a: The main purpose of the leading time of K_a sample periods is to improve system stability margins introducing a leading action on the controller at the periodic frequencies so as to compensate the delay of the process $P(z)$ in Fig. 7(b). The design of this parameter is based on the number of delay samples K_a, which better approximates the delay of transfer function $P(z)$ at the harmonic frequencies. A good approximation of the phase of $P(z)$ is at about three sample periods, that is, $K_a = 3$. Fig. 8(a) reports the magnitude of the repetitive control output, highlighting the frequencies where the repetitive control gives high gain, and the magnitude of the PI control output, taking into account that the PI regulator has been designed to achieve a bandwidth of 3900 rad/s with a phase margin of 0.6 rad. Fig. 8(b) reports the comparison between the phase of $P(z)$ and the phase of a delay equal to three sample periods, in this range of interest.

The approximation is quite good, at least for frequencies above 300 Hz. Indeed, the high-frequency range is the most critical since in the low-frequency range the PI control has a higher gain and it overrides the repetitive control.

2) Design of Parameter K_{REP}: The repetitive control main task is the compensation of the steady-state error harmonics. The gain of the repetitive control outside those frequencies should be kept quite smaller compared to the gain of the PI control, so that the slow dynamics of the repetitive part does not influence the transient behavior of the drive. The gain of the PI control coincides with K_p around the PI bandwidth limit frequencies and above. Then, a choice of K_{REP} in the range of $K_p/5 \div K_p/20$ is recommended. A refined choice of K_{REP} comes from stability analysis, as reported hereafter.

3) Stability Analysis: The stability analysis will be first performed on a case study, by assuming a speed of 30% of the nominal speed, which is equivalent to $M = 30$. Then, the analysis will be extended to different speeds and M. Actually, the analysis of the stability under varying motor speed is quite involved, since parameters M and T_c are functions of the motor speed, which is theoretically a function of the repetitive control. In practice, however, it is possible to neglect the contribution of repetitive control output $u^*_{q\mathrm{REP}}$ to the motor speed variations, thus simplifying the stability analysis. Using the procedure outlined in the previous paragraphs, the real and imaginary part of the function $H(e^{j\omega T_c})$ have been computed, with ω ranging from 0 up to π/T_c (Nyquist angular frequency), and using $M = 30, K_a = 3$, and $K_{\mathrm{REP}} = 0.1\,K_p$. As shown in Fig. 9, condition (11) is satisfied and, consequently, the stability of the proposed control is ensured.

In order to verify the feasible range of parameters K_a and K_{REP}, the stability has been verified for different values, as reported in Fig. 10. The system stability was asserted when the amplitude of the roots of the denominator in (10) were less than one, which yields to a theoretical stable system. Nevertheless, the stable region shown in Fig. 10 has been verified by some approximations also in the experimental setup. Note that the proposed design places the parameters far away from the instability region. Finally, the stability limits have been recalculated for different motor speeds or, equivalently, by varying M. It has been found that the stability boundaries do not change significantly as the motor speed is changed. For example, there are no significant differences in Fig. 10 when the motor speed is changed from 0.1 up to three times the speed assumed in the previous analysis. Robustness of the proposed solution with respect to parameter variations in the motor inductance are not

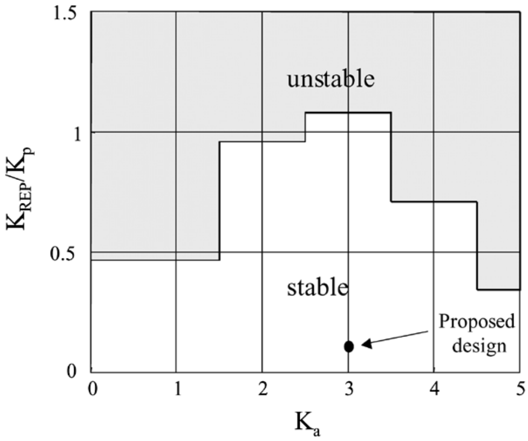

Fig. 10. Stability limits of the proposed solution with $M = 30$.

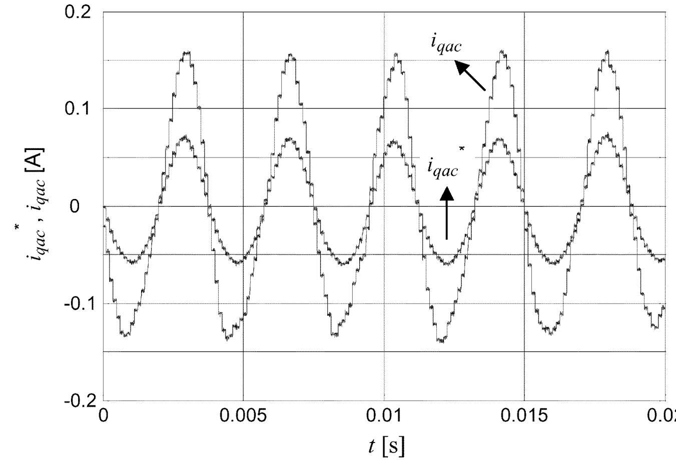

Fig. 11. Current tracking with conventional PI control.

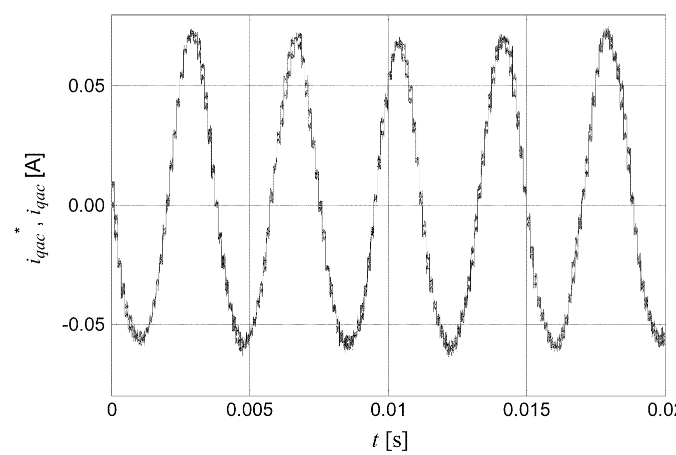

Fig. 12. Current tracking with repetitive control.

considered here since the motor is with a surface-mounted PM, so that the equivalent (augmented) airgap prevents inductances from saturation.

IV. Experimental Results

The proposed drive has first been simulated, and then implemented on a DSP-based laboratory prototype. An induction motor driven by a high dynamic standard drive has been used as variable load, while a custom inverter, equipped with a DSP board and interface facilities with a PC has been used to drive the PMSM. The repetitive algorithm, the speed and current control loops, the SVM timing calculations, and dead-time compensation have been implemented on a Texas Instruments TMS320C31 floating-point DSP, with a 33.3-ns instruction cycle. All software procedures are written in C language and optimized for shortest execution time. Digital input/output (I/O) management and switching pattern generation are handled by a slave fixed-point TMS320P14 DSP, with 160-ns instruction cycle, featuring 16 individual bit-selectable I/O and six PWM channels, with a period resolution of 80 ns. Control software routines for the slave processor have been written in Assembly language, for a closer link with the embedded hardware peripherals. At first, for comparison, the standard PI current controller has been tested alone. As expected, due to the limited bandwidth, the PI does not track the reference properly. As a consequence, a steady-state current error appears, as documented in Fig. 11, for a given torque reference of 40% of the nominal value. For the sake of graphical rendering, Fig. 11 and all the others figures that deal with q-axis current plots, show only the comparison between the ac current components. Actually, due to the PI action, the dc component i_{qdc}^* is always perfectly matched by the current control.

The actual i_{qac} shows the effects of both the reference i_{qac}^* and the back-EMF e_q, which acts as a disturbance input. It is easy to recognize that the latter yields a phase advance in the response that compensates, almost completely, the PI phase lag, at the selected test frequency $\omega_{\mathrm{rep}} = 1675$ rad/s [18], [19].

Under the same torque demand, the PI controller has been paralleled by the repetitive control, according to the scheme of Fig. 6. Fig. 12 reports the behavior of the resulting current controller, showing a perfect match between the actual q-axis current i_q and its reference i_q^*.

With reference to (9), the result of Fig. 12 has been obtained by setting $M = 30, K_a = 3, K_{\mathrm{REP}} = 3, \omega_m = 0.3\ \Omega_n, \tau = 40\%\ \tau_N$. Both Figs. 11 and 12 refer to a steady-state analysis of the drive proposed in Fig. 6.

The dynamic behavior has been investigated by starting the current control with the PI only, and then studying the current transient that follows the enabling of the repetitive algorithm. Fig. 13 reports the reference and actual current, respectively, and Fig. 14 the current error. Note that the repetitive control requires slightly more than 100 ms to settle down, or equivalently around 27 times the repetitive period (T_{rep}). This is the best tradeoff between stability and response time that was achievable with the given experimental setup, taking into account different speed and loading conditions. In order to quantify the effectiveness of the proposed solution, the motor torque has been estimated starting from electrical variables [20] and by neglecting both the torque terms due to rotor anisotrophy (since the motor has surface-mounted PM) and the cogging torque, which was unaffected by both PI and REP control strategies. Finally, Fig. 15 reports the estimated motor torque, with a reference $\tau^* = 40\%\tau_n$. Note that with the standard PI regulator the superposed peak ripple is about 6%, while it is almost negligible with the proposed solution.

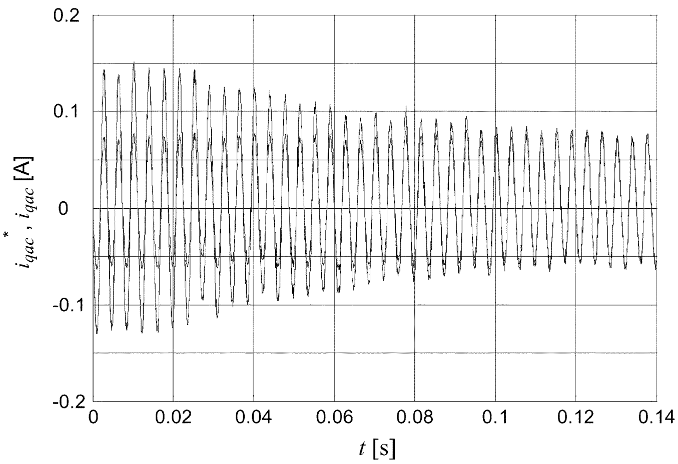

Fig. 13. Current tracking transient after repetitive control enable.

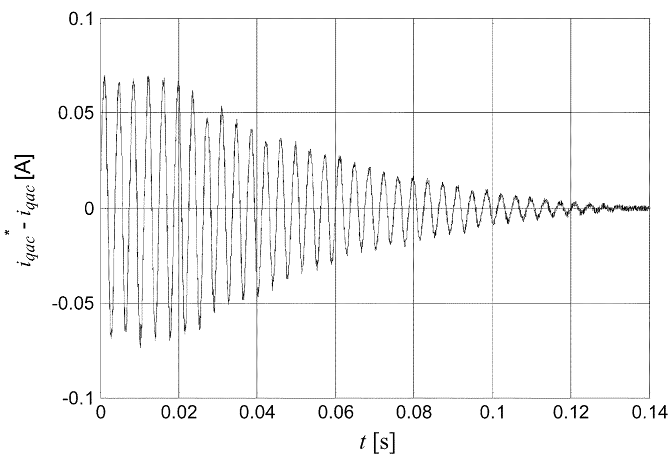

Fig. 14. Current error transient after repetitive control enable.

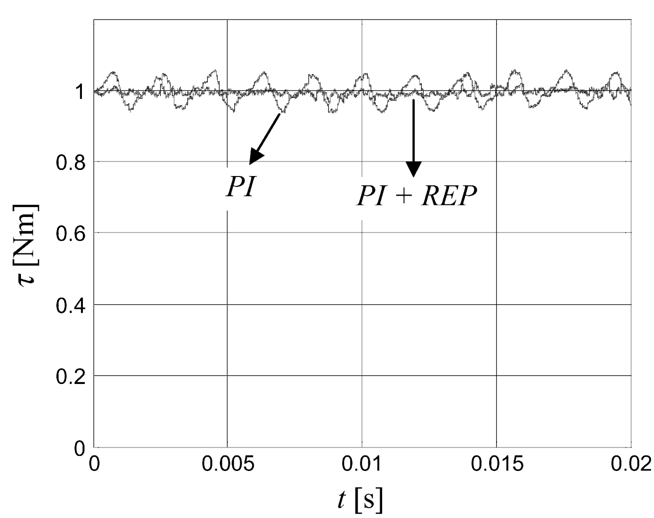

Fig. 15. Estimated motor torque, PI, and proposed solution under comparison.

V. Further Design Hints

A. Speed Measurement Accuracy

It is important to highlight that (8) expresses a stringent condition for a correct tracking of the current reference. As shown in Fig. 8(a) and owing to the intrinsic property of the repetitive algorithm, the frequency response is heavily selective and the gain quickly drops around the fundamental frequency and its multiple. Nevertheless, a fairly small frequency shift around the fundamental component, say less then 0.5%, yields limited gain variations and the control strategy is still effective. From a different perspective, the mechanical speed has to be tightly synchronized with an integer number (M) of current samples in order to avoid any loss of selectivity in the repetitive control. The absolute position was acquired by a resolver and a resolver-to-digital converter with N-bit selectable resolution ($N = 12$, 14, 16). The speed was calculated according to

$$\omega_m = \frac{1}{6T_c F_{\rm ds}}[\vartheta_k - \vartheta_{k-3} + 3(\vartheta_{k-1} - \vartheta_{k-2})] \qquad (12)$$

where position samples are taken every $T_c F_{\rm ds}$, being $F_{\rm ds} = 20$ the selected downsampling factor. From (12), with N bit resolution, the worst-case absolute speed resolution can be easily estimated

$$d\omega_m = \frac{2\pi}{2^N}\frac{8}{6\,T_c F_{\rm ds}}. \qquad (13)$$

As a rule of thumb, the lowest speed at which the repetitive action should be still effective can be derived from the Bode diagram of repetitive and PI controllers, reported in Fig. 8(a). A lower synchronous speed yields to a left shift of the REP curve. It is proposed that the minimum speed to be considered is the fundamental angular frequency $\omega_{e,\min}$ at which the REP and PI gains coincide. At a lower speed, the PI action is dominant and the repetitive control can be deactivated. In the present case [Fig. 8(a)], it was $\omega_{e,\min} = 180$ rad/s and then, according to (8), $\omega_{m,\min} = 10$ rad/s. The required resolution in the speed measurement, that has to be achieved in the whole operating speed range, has then to be verified at its lower boundary, as expressed by the following condition:

$$\frac{d\omega_m}{\omega_{m,\min}} = \frac{2\pi}{2^N}\frac{8}{6\,T_c F_{\rm ds}\omega_{m,\min}} \le 0.005. \qquad (14)$$

In the present work, the condition (14) is met by a resolver-to-digital resolution $N = 16$. However, since (13) represents the worst case and taking into account that the average speed is kept equal to its reference value even in the presence of lower resolution because of the integral action on the speed loop, it has been verified by simulation and experimental test that the resolver-to-digital resolution can be much lower than condition (14). In particular, the experimental results on the laboratory prototype have shown good current tracking even with $N = 12$.

B. Selection of Sampling Time and the Number of Samples

A stiff requirement of repetitive control is that the current sampling remains somehow synchronized with the periodic current reference, so that a reference period contains always an integer number M of samplings T_c, and the controller structure of Fig. 5(c) holds. After any speed reference variation, at steady-state, both T_c and M have to be properly modified in order to meet the condition (8). In the laboratory prototype, the resolution on T_c was $\Delta T_c = 80$ ns. Hence, the reference speed

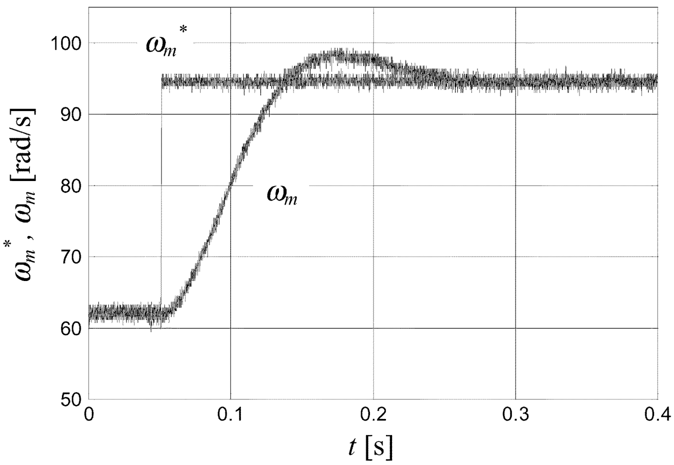

Fig. 16. Speed small step response of the proposed drive.

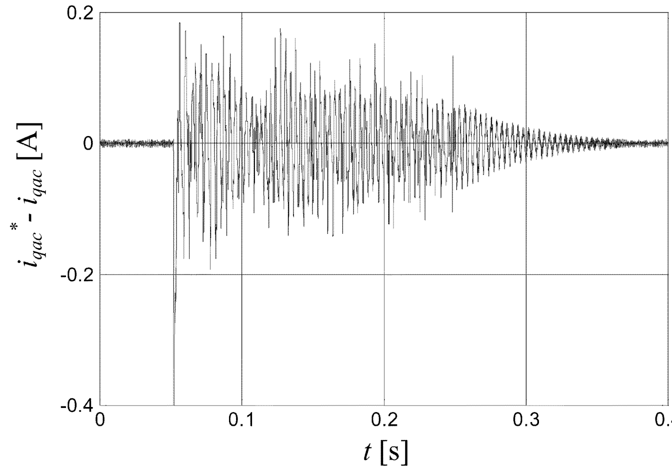

Fig. 17. q-axis current error behavior during a speed transient.

ω_m^* can vary only by finite steps $\Delta\omega_m$, according to the following equation:

$$\Delta\omega_m = \frac{\pi\Delta T_c}{3pM(T_c - \Delta T_c)T_c} \approx \frac{\pi}{3pM}\left(-\frac{1}{T_C^2}\right)\Delta T_C. \quad (15)$$

The minimum speed step, compatible with the available sampling time resolution, is given by

$$\frac{\Delta\omega_m^*}{\omega_m^*} = \frac{\Delta T_c}{(T_c - \Delta T_c)} \approx \frac{\Delta T_c}{T_c} \quad (16)$$

which leads to a speed resolution of $0.06 \div 0.08\%$ for sampling times included in the range $100 \div 125$ μs.

For any given speed, both M and T_c have to be selected. It preferable to first choose the parameter M as

$$M = \text{ round}\left(\frac{2\pi}{\omega_m^* phT_c}\right) \quad (17)$$

and then

$$T_c = \frac{2\pi}{\omega_m^* phM} \quad (18)$$

where the function $\text{round}(x)$ rounds the element x to the nearest integer. In the computation of (17), it is necessary to start always from the same T_c, to avoid misconvergence of the algorithm. Within the range $\omega_{m,\min} \leq \omega_m \leq \Omega_N$, which was the operative range of the repetitive control in this work, M varies from 9 to 279 and T_c oscillates between 119 μs and 131 μs. It is worth to highlight that the M upper limit fixes the memory requirement, that could be critical for low-cost microprocessors. Note, however, that a down-sampling strategy on the repetitive control could be adopted at low speed to avoid the increase of memory requirement. In any case, it is advisable to avoid excessive T_c variations, since the PI dynamics can be compromised, leading also to other undesired side effects, such as high-frequency current ripple or increased switching losses. Fig. 16 reports that the drive behavior during a speed transient from $\omega_m = 0.2\,\omega_N(M = 45)$ to $\omega_m = 0.3\,\omega_N(M = 30)$, both with $T_c = 125$ μs. The related current error is shown in Fig. 17.

At first, the behavior is dominated by PI control, but as soon as the speed has reached the new steady-state, the REP output limitation is released and its gradual action yields to a practically null error in about 0.1 s, confirming the effectiveness of the proposed control strategy even during speed transients.

VI. Conclusion

The paper has proposed a repetitive-based control for torque-ripple reduction in PM motor drives with nonsinusoidal back-EMF. A smooth torque production is obtained by a proper q-axis current reference, based on the specific motor back-EMF. The reference, which contains an alternating component superposed to the conventional dc component, is tracked by the parallel of a PI and a repetitive controller. The proposed repetitive controller, specifically designed for this application, is committed to the reduction of the high-frequency periodic current error, which cannot be compensated by the PI due to its limited bandwidth. The result, supported by both simulations and experimental tests, is an almost perfect current tracking in the whole operating range of the drive, and a consequently absolutely smooth torque production. The proposed drive, while requiring good computational power and memory availability, seems to fit all of those industrial applications which do not bear any torque-ripple, such as spindle drives and accurate positioning systems.

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References

[1] L. Springob and J. Holtz, "High-bandwidth current control for torque-ripple compensation in PM synchronous machines," *IEEE Trans. Ind. Electron.*, vol. 45, no. 5, pp. 713–720, Oct. 1998.

[2] S. Bolognani, L. Tubiana, and M. Zigliotto, "Sensorless control of PM synchronous motors with nonsinusoidal back-EMF for home appliances," in *Proc. IEEE Int. Machines Drives Conf. (IEMDC'03)*, Madison, WI, 2003, pp. 1882–1888.

[3] T. Jahns and W. Soong, "Pulsating torque minimization techniques for permanent magnet AC motor drives—A review," *IEEE Trans. Ind. Electron.*, vol. 43, no. 2, pp. 321–330, Apr. 1996.
[4] P. L. Chapman, S. D. Sudhoff, and C. A. Whitcomb, "Optimal current control strategies for surface-mounted permanent-magnet synchronous machine drives," *IEEE Trans. Electromag. Compat.*, vol. 14, no. 4, pp. 1043–1050, Dec. 1999.
[5] B. A. Francis and W. M. Wonham, "The internal model principle of control theory," *Automatica*, vol. 12, pp. 457–465, 1976.
[6] G. Hillestrom, "Adaptive suppression of vibrations—A repetitive control approach," *IEEE Trans. Contr. Syst. Technol.*, vol. 4, no. 1, pp. 72–78, Jan. 1996.
[7] G. Hillerström and K. Walgama, "Repetitive control theory and applications—A survey," in *Proc. 13th IFAC World Congr.*, vol. D, San Francisco, CA, Jul. 1996, pp. 1–6.
[8] T. Inoue, "Practical repetitive control system design," in *Proc. 29th IEEE Conf. Decision Control*, Honolulu, HI, Dec. 1990, pp. 1673–1678.
[9] J. Ghosh and B. Paden, "Nonlinear repetitive control," *IEEE Trans. Automat. Contr.*, vol. 45, no. 5, pp. 949–954, May 2000.
[10] M. Yamada, Z. Riadh, and Y. Funabashi, "Design of discrete-time repetitive control system for pole placement and application," *IEEE/ASME Trans. Mechatron.*, vol. 4, no. 2, pp. 110–118, Jun. 1999.
[11] J. H. Moon, M. N. Lee, and M. J. Chung, "Repetitive control for the track-following servo system of an optical disk drive," *IEEE Trans. Contr. Syst. Technol.*, vol. 6, no. 6, pp. 663–670, Sep. 1998.
[12] K. K. Chew and M. Tomizuka, "Digital control of repetitive errors in disk drive systems," *IEEE Contr. Syst. Mag.*, vol. 10, no. 1, pp. 16–20, Jan. 1990.
[13] C. Rech, H. Pinheiro, H. A. Grundling, H. L. Hey, and J. R. Pinheiro, "Analysis and design of a repetitive predictive-PID controller for PWM inverters," in *Proc. IEEE 32nd Annu. Power Electronics Specialists Conf. (PESC'01)*, vol. 2, Vancouver, BC, Canada, 2001, pp. 986–991.
[14] Y. Y. Tzou, S.-L. Jung, and H. C. Yeh, "Adaptive repetitive control of PWM inverters for very low THD AC-voltage regulation with unknown loads," *IEEE Trans. Power Electron.*, vol. 14, no. 5, pp. 973–981, Sep. 1999.
[15] P. Mattavelli and F. Marafao, "Selective active filters using repetitive control techniques," *IEEE Trans. Ind. Electron.*, vol. 51, no. 5, pp. 1018–1024, Oct. 2004.
[16] K. Zhang, Y. Kang, J. Xiong, and J. Chen, "Direct repetitive control of SPWM inverters for UPS purpose," *IEEE Trans. Power Electron.*, vol. 18, no. 3, pp. 784–792, May 2003.
[17] K. Zhou and D. Wang, "Digital repetitive controlled three-phase PWM rectifier," *IEEE Trans. Power Electron.*, vol. 18, no. 1, pp. 309–316, Jan. 2003.
[18] T. Sebastian and V. Gangla, "Analysis of induced EMF waveforms and torque-ripple in a brushless permanent magnet machine," *IEEE Trans. Ind. Appl.*, vol. 32, no. 1, pp. 195–200, Jan./Feb. 1996.
[19] M. Yoshida, Y. Murai, and M. Takada, "Noise reduction by torque-ripple suppression in brushless DC motor," in *Proc. IEEE Power Electronics Specialists Conf. (PESC'98)*, Fukuoka, Japan, 1998, pp. 1397–1401.
[20] D. C. White and H. H. Woodson, *Electromechanical Energy Conversion*. New York: Wiley, 1959.

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